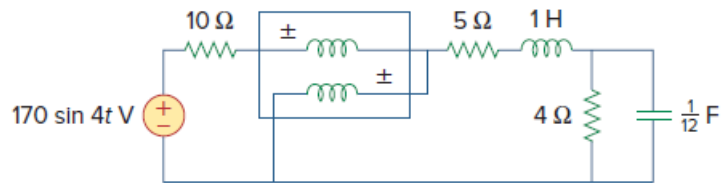


Problem

Find the wattmeter reading of the circuit shown in Fig. 11.93.

Figure 11.93:



Step-by-step solution

Step 1 of 7

Refer to Figure 11.93 in the text book.

Consider the value of the voltage source v_s .

$$v_s = 170 \sin 4t \text{ V}$$

Compare with $v_m \sin \omega t$.

Therefore, the value of v_m and ω are 170 V and 4 rad/sec respectively.

Convert the value of the voltage to RMS value and represent in phasor form.

$$\begin{aligned} V_s &= \frac{170}{\sqrt{2}} \angle 0^\circ \\ &= 120.21 \angle 0^\circ \text{ V rms} \end{aligned}$$

Therefore, the value of the source voltage in phasor form is, $120.21 \angle 0^\circ \text{ V rms}$.

Step 2 of 7

Calculate the value of the reactance due to inductance.

$$X_L = j\omega L$$

Substitute 4 rad/sec for ω and 1 H for L .

$$\begin{aligned} X_L &= j(4)(1) \\ &= j4 \Omega \end{aligned}$$

Therefore, the value of the reactance X_L due to inductance is, $j4 \Omega$.

Step 3 of 7

Calculate the value of the reactance due to capacitance.

$$X_c = \frac{1}{j\omega C}$$

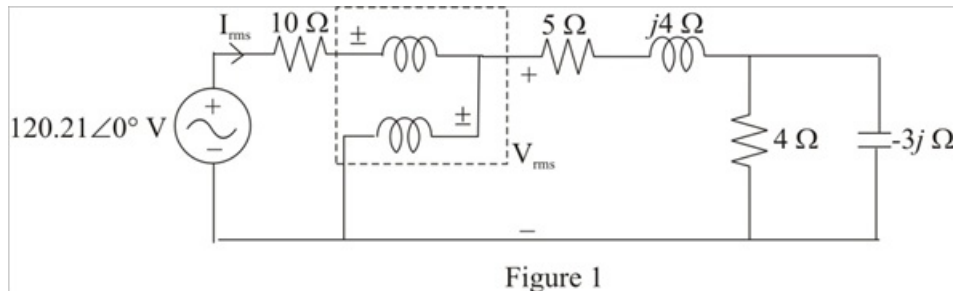
Substitute 4 rad/sec for ω and $\frac{1}{12}$ F for C .

$$\begin{aligned} X_c &= \frac{1}{j(4)\left(\frac{1}{12}\right)} \\ &= -j3 \Omega \end{aligned}$$

Therefore, the value of the reactance X_c due to capacitance is, $-j3 \Omega$.

Step 4 of 7

Consider the circuit diagram shown in Figure 1.



Step 5 of 7

Calculate the value of the current I_{rms} passing in the circuit.

$$\begin{aligned} I_{rms} &= \frac{120.28\angle 0^\circ}{10 + 5 + j4 + (4 \parallel -j3)} \\ &= \frac{120.28\angle 0^\circ}{15 + j4 + \frac{(4)(-j3)}{4 - j3}} \\ &= \frac{120.28\angle 0^\circ}{15 + j4 + \frac{-j12}{4 - j3}} \\ &= \frac{120.28\angle 0^\circ}{15 + j4 + 1.44 - j1.92} \\ I_{rms} &= \frac{120.28\angle 0^\circ}{16.44 + j2.08} \\ &= \frac{120.28\angle 0^\circ}{16.57\angle 7.21^\circ} \\ &= 7.26\angle -7.21^\circ \text{ A rms} \end{aligned}$$

Therefore, the value of the current I_{rms} is $7.26\angle -7.21^\circ$ A rms.

Step 6 of 7

Calculate the value of the voltage V_{rms} across the load.

$$V_{\text{rms}} = I_{\text{rms}} (5 + j4 + (4 \parallel -j3))$$

Substitute $7.26 \angle -7.21^\circ \text{ A rms}$ for I_{rms} .

$$\begin{aligned} V_{\text{rms}} &= (7.26 \angle -7.21^\circ) \left(5 + j4 + \frac{(4)(-j3)}{4 - j3} \right) \\ &= (7.26 \angle -7.21^\circ) \left(5 + j4 + \frac{-j12}{4 - j3} \right) \\ &= (7.26 \angle -7.21^\circ) (5 + j4 + 1.44 - j1.92) \\ &= (7.26 \angle -7.21^\circ) (6.44 + j2.08) \\ V_{\text{rms}} &= (7.26 \angle -7.21^\circ) (6.768 \angle 17.9^\circ) \\ &= 49.14 \angle 10.69^\circ \text{ V rms} \end{aligned}$$

Therefore, the value of the voltage V_{rms} across the load is $49.14 \angle 10.69^\circ \text{ V rms}$.

Step 7 of 7

Notice that Wattmeter reads only the average power absorbed by the load.

Calculate the value of the average power absorbed by the load.

$$P = V_{\text{rms}} I_{\text{rms}} \cos(\theta_v - \theta_i)$$

Substitute 49.14 V rms for V_{rms} , 10.69° for θ_v , -7.21° for θ_i and 7.26 A rms for I_{rms} .

$$\begin{aligned} P &= (49.14)(7.26) \cos(10.69^\circ - (-7.29^\circ)) \\ &= (356.75) \cos(10.69^\circ + 7.29^\circ) \\ &= (356.75) \cos(17.98^\circ) \\ &= (356.75)(0.9512) \end{aligned}$$

$$P = 339.34 \text{ W}$$

Therefore, the value of the wattmeter reading is, 339.34 W .